

FIITJEE

ALL INDIA TEST SERIES

FULL TEST – XV

JEE (Main)-2022

TEST DATE: 15-06-2022

Time Allotted: 3 Hours

Maximum Marks: 300

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts**.
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B** : Do any five questions out of 10 Questions.

Section-A (01 – 20, 31 – 50, 61 – 80) contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Section-B (21 – 30, 51 – 60, 81 – 90) contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Physics

PART – A

SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

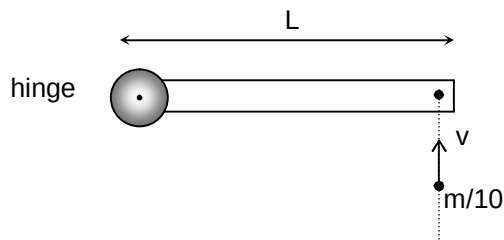
- The kinetic energy K of a particle moving along a circle of radius R depends upon the distance moved from rest s , as $K = as^2$. The force acting on the particle is (a is a positive constant)
 - $2a \frac{s^2}{R}$
 - $2as \left[1 + \frac{s^2}{R^2} \right]^{1/2}$
 - $2as$
 - $2a$
- In which of the following instrument used in the lab there exists an error of random category known as back lash error
 - (i) Screw gauge (ii) Micrometer (iii) Vernier callipers
 - (i) and (ii) only
 - (i) and (iii) only
 - (i) only
 - three
- A body cools down from 50°C to 45°C in 5 min and 45°C to 40°C in 8 min. Find the temperature of the surrounding. Assume Newton's law of cooling is valid.
 - 34°C
 - 24°C
 - 39°C
 - 20°C
- When a belt moves horizontally at a constant speed of 1.5ms^{-1} , gravel is falling on it at 5kgs^{-1} continuously and sticks to it, then the extra power needed to drive the belt is
 - 11.25 W
 - 37.5 W
 - 7.5 W
 - 0.75 W
- Electrons with de Broglie wavelength λ fall on the target in an X-ray tube. The cut-off wavelength of the emitted X-rays is
 - $\frac{2mc\lambda^2}{h}$
 - $\frac{2h}{mc}$
 - $\frac{2m^2c^2\lambda^3}{h^2}$
 - λ

6. A block of mass $m = 3\text{ kg}$ resting on a horizontal frictionless floor is horizontally struck by a 9 N force that acts for 0.02 sec . After 3 sec it receives a second blow of force 9 N but in opposite direction which acts for 0.01 sec . The speed of the body after 10 sec . is:

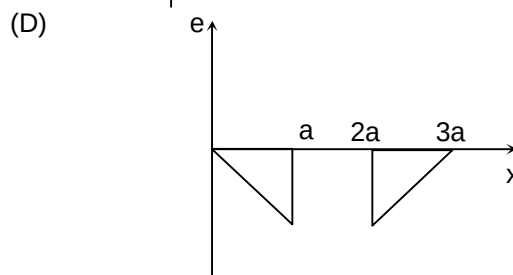
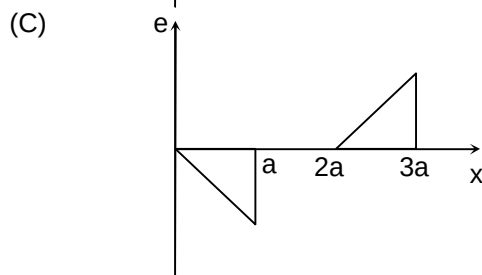
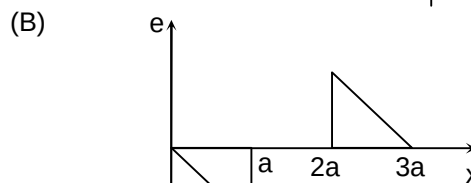
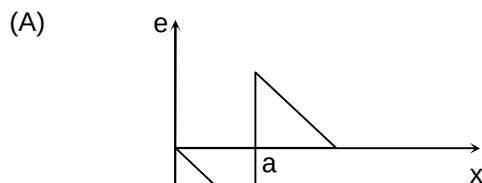
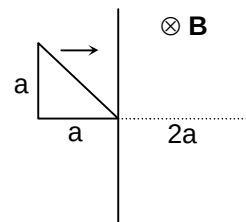
(A) 0
 (B) 3 cm/sec
 (C) 10^4 m/sec
 (D) 10^5 m/sec

7. A thin uniform rod of mass ' m ' of length ' L ' hinged at one end. This rod is maintained in horizontal position by colliding very tiny balls of each mass $m/10$ completely elastically 10 times per sec striking on the opposite end as shown in figure. Find the speed of the ball. (Take $g = 10\text{ m/s}^2$)

(A) 10 m/sec
 (B) 2.5 m/sec
 (C) 5 m/sec
 (D) 20 m/sec



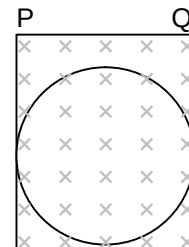
8. A right angled triangular loop travelling with constant velocity (as shown in the figure) enters in uniform magnetic field (at right angle to the boundary of the field) directed into the paper. Draw the graph between induced emf e and the distance along the perpendicular to the boundary of the field (say X) along which loop moves



9. A coil is in the form of equilateral triangle of side ' a '. The current in the coil is I . The magnetic field at a distance ' a ' equidistant from all the vertices will be

(A) $\frac{\mu_0 I}{2\sqrt{3}\pi a}$
 (B) $\frac{\sqrt{3}\mu_0 I}{2\pi a}$
 (C) $\frac{\mu_0 I}{\sqrt{3}\pi a}$
 (D) $\frac{\mu_0 I}{4\sqrt{3}\pi a}$

10. A vertical ring of radius r and resistance R falls vertically. It is in contact with two vertical rails which are joined at the top. The rails are without friction and resistance. There is a horizontal uniform magnetic field of magnitude B perpendicular to the plane of the ring and the rails. When the speed of the ring is v , the current in the section PQ is

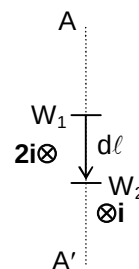


- (A) zero
(B) $\frac{2Brv}{R}$
(C) $\frac{4Brv}{R}$
(D) $\frac{8Brv}{R}$

11. Two spheres each of mass $8\sqrt{3}$ kg and radius 5 cm lie in contact inside a hemispherical bowl of radius 40 cm. There is no friction between any of the contact surface. The contact force exerted by one sphere on another is (Take $g = 10 \text{ m/s}^2$)

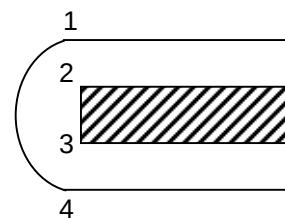
- (A) 10 N
(B) 20 N
(C) 30 N
(D) 40 N

12. Two wires W_1 & W_2 carry current ' $2i$ ' and i respectively directed into the plane of paper as shown in figure. AA' is line at some distance from wire. Let $\vec{B}$ be the net magnetic field due to wires. The magnitude of $\int_{-\infty}^{\infty} \vec{B} \cdot d\vec{\ell}$ will be equal to



- (A) $\mu_0 i / 2$
(B) $\mu_0 i / 4$
(C) $\mu_0 i / 8$
(D) zero

13. Four identical plates 1, 2, 3 and 4 are placed parallel to each other at equal distance as shown in the figure. Plates 1 and 4 are joined together and the space between 2 and 3 is filled with dielectric of dielectric constant $k = 2$. The capacitance of the system between 1 and 3 is C_1 and between plates 2 and 4 is C_2 . The ratio C_1/C_2 is:



- (A) $\frac{5}{3}$
(B) 1
(C) $\frac{3}{5}$
(D) $\frac{5}{7}$

14. An arc of radius r carries charge. The linear density of charge is λ and the arc subtends an angle $\frac{\pi}{3}$ at the centre. What is electric potential at the centre?

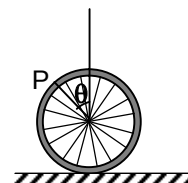
- (A) $\frac{\lambda}{4\epsilon_0}$

- (B) $\frac{\lambda}{8\epsilon_0}$
 (C) $\frac{\lambda}{12\epsilon_0}$
 (D) $\frac{\lambda}{16\epsilon_0}$

15. A sphere, a cube and a thin circular plate all made of same material and having the same mass are initially heated to a temperature of 200°C . Which of these objects will cool fastest and which will cool slowest when left in air at room temperature, respectively?
 (A) Sphere, cube
 (B) Cube, thin circular plate
 (C) Thin circular plate, sphere
 (D) All will cool at the same rate

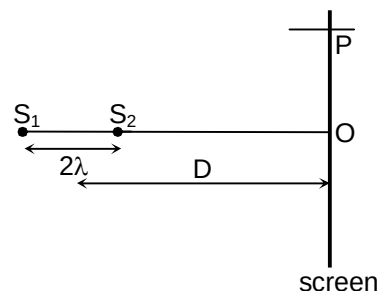
16. A wheel is rolling straight on ground without slipping. If the centre of mass of the wheel has speed v , then instantaneous velocity of a point P on the rim, defined by angle θ with vertical, relative to the ground will be

- (A) $v \cos\left(\frac{1}{2}\theta\right)$
 (B) $2v \cos\left(\frac{1}{2}\theta\right)$
 (C) $v \sin\left(\frac{1}{2}\theta\right)$
 (D) $v \sin(\theta)$

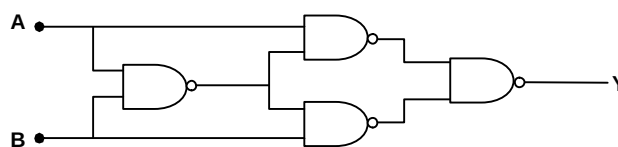


17. Two sources S_1 and S_2 emitting coherent light waves of wavelength λ in the same phase are situated as shown. The minimum distance OP , so that the light intensity detected at P is equal to that at O is ($D \gg \lambda$)

- (A) $D\sqrt{2}$
 (B) $D/2$
 (C) $D\sqrt{3}$
 (D) $D/\sqrt{3}$



18. Truth table for system of four NAND gates as shown in figure is :



- (A)
- | A | B | Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

- (B)
- | A | B | Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

(C)	A	B	Y
	0	0	1
	0	1	1
	1	0	0
	1	1	0

(D)	A	B	Y
	0	0	1
	0	1	0
	1	0	0
	1	1	1

19. The activity of a sample of radioactive material depends on
 (A) its temperature
 (B) its chemical composition
 (C) number of atoms present
 (D) all of these factors
20. A charged particle enters in uniform magnetic field with velocity at an angle of 45° with magnetic field. The pitch of the helical path followed by the particle is P . The radius of the helix will be
 (A) $\frac{P}{\sqrt{2}\pi}$
 (B) $\sqrt{2}P$
 (C) $\frac{P}{2\pi}$
 (D) $\frac{\sqrt{2}P}{\pi}$

SECTION – B (Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

21. The near vision of an average person is 25 cm. To view an object with an angular magnification of 10 the power of the microscope is k diopters. Find the value of k .
22. In a uniform gravitational field g , which is acting vertically downward, a ball is thrown at an angle with horizontal such that the initial (immediately after projection) radius of curvature is 8 times the minimum radius of curvature. If angle of projection is 150° , then find the value of θ .
23. The minimum force required along the plane to start pushing a body up a rough (frictional coefficient μ) inclined plane is F_1 while the maximum force along the plane needed to prevent it from sliding down is F_2 . If the inclined plane makes an angle θ from the horizontal such that $\tan\theta = 2\mu$, then the ratio F_1 / F_2 is
24. A 3 kg sphere makes a perfectly inelastic head-on collision with a second sphere initially at rest. The composite system moves with a speed equal to one third of the original speed of the 3 kg sphere. Find ratio of initial and final kinetic energy of system.
25. A series combination of n_1 capacitors, each of value C_1 , is charged by a source of potential difference 4V. When another parallel combination of n_2 capacitors, each of value C_2 , is charged by a source of potential difference V , it has the same (total energy) stored in it as the first combination has. The value of C_2 in terms of C_1 , is $\frac{kC_1}{n_1n_2}$. Find the value of k .

26. The length of a cylinder is measured with a metre rod having least count of 0.1 cm. Its diameter is measured with a Vernier Callipers with Vernier constant 0.01 cm. Given that the length of the cylinder is 5.0 cm and its radius is 2.00 cm. The percentage error in the calculated volume is x . find x
27. Water is boiled in a rectangular steel tank of thickness 2 cm by a constant temperature furnace. Due to vaporization, water level falls at a steady rate of 1 cm in 9 minutes. If the temperature of furnace is $n \times 22^\circ\text{C}$. Find n . Given K for steel = 0.2cgs units, $L_{\text{vapour}} = 540 \text{ Cal/gm}$.
28. A series L – R a.c circuit with power factor $\frac{1}{\sqrt{2}}$ has source voltage 8V. What is the peak voltage (in V) across the inductor?
29. One kg of a diatomic gas is at a pressure of $8 \times 10^4 \text{ N/m}^2$. The density of the gas is 4 kg/m^3 . The energy of the gas due to its thermal motion is $p \times 10^4 \text{ J}$. Find p . (neglect vibrational energy)
30. A cubical block is floating in a liquid with half of its volume immersed in the liquid. When the whole system accelerates upward with a net acceleration of $g/2$, then find the fraction of volume immersed in the liquid.

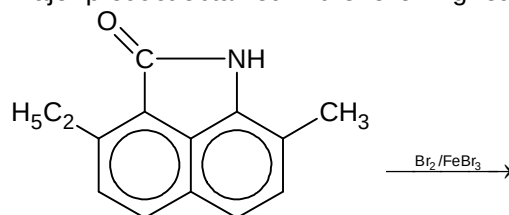
Chemistry

PART – B

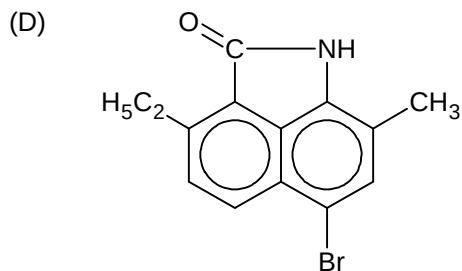
SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

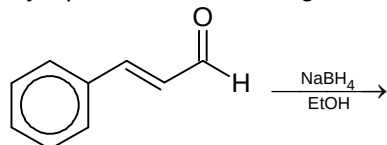
31. The element with atomic number 56 belongs to
 (A) s-Block element
 (B) p-Block element
 (C) d-Block element
 (D) f-Block element
32. Which of the following is yellow colour sulphide?
 (A) ZnS
 (B) HgS
 (C) CdS
 (D) Bi₂S₃
33. Major product obtained in the following reaction



- (A)
- (B)
- (C)



34. Major product obtained in given reaction is



- (A)
- (B)
- (C)
- (D)

35. Which of the following alcohol will be oxidized to acid using Br_2 / KOH , followed by work up?

- (A) $\text{CH}_3 - \text{OH}$
- (B)
- (C)
- (D)

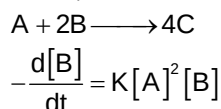
36. If the uncertainty in position is $\frac{1}{4}$ th of uncertainty in momentum then the minimum value of uncertainty in velocity will be

- (A) $\sqrt{\frac{h}{2\pi}}$
 (B) $\frac{1}{m} \sqrt{\frac{h}{2\pi}}$
 (C) $\frac{1}{m} \sqrt{\frac{h}{\pi}}$
 (D) $\sqrt{\frac{h}{\pi}}$

37. A 100 ml solution is prepared by using x ml 0.2 M NaOH and rest 0.1 M H_2SO_4 , what will be the value of x among the following option in which maximum rise in temperature observe

- (A) 30 ml
 (B) 70 ml
 (C) 40 ml
 (D) 49 ml

38. For a reaction,



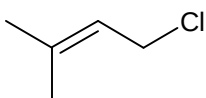
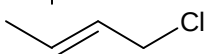
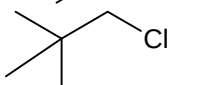
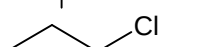
The expression for $\frac{d[C]}{dt}$ will be

- (A) $2K[A][B]^2$
 (B) $K[A]^2[B]$
 (C) $\frac{1}{2}K[A]^2[B]$
 (D) $2K[A]^2[B]$

39. The linear molecule is/are

- (I) $SnCl_2$ (II) NCO^-
 (III) NO_2^+ (IV) N_3^-
 (A) I, II & III
 (B) I & IV
 (C) II & IV
 (D) All of these

40. Which of the following substrate has highest rate for S_N1 reaction?

- (A) 
 (B) 
 (C) 
 (D) 

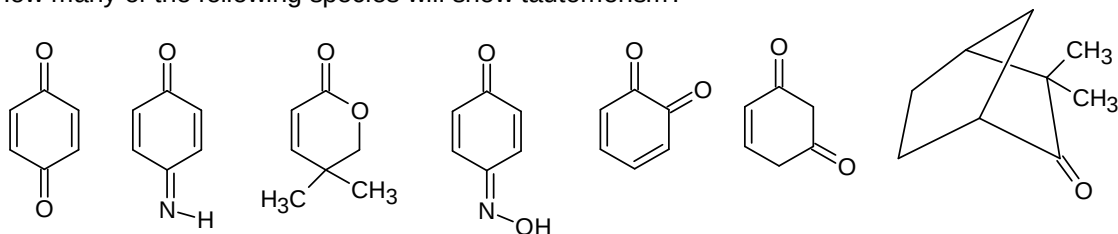
41. The common ion effect shown by
(A) $\text{NaCl} + \text{NaOH}$
(B) $\text{KCl} + \text{HCl}$
(C) $\text{H}_2\text{O} + \text{HCl}$
(D) $\text{BaCl}_2 + \text{Ba}(\text{NO}_3)_2$
42. The correct order of relative acidic strength is
(A) $\text{CH}_3\text{COOH} > \text{HCN} > \text{H}_2\text{O} > \text{C}_2\text{H}_5\text{OH}$
(B) $\text{CH}_3\text{COOH} > \text{H}_2\text{O} > \text{C}_2\text{H}_5\text{OH} > \text{HCN}$
(C) $\text{HCN} > \text{CH}_3\text{COOH} > \text{H}_2\text{O} > \text{C}_2\text{H}_5\text{OH}$
(D) $\text{C}_2\text{H}_5\text{OH} < \text{HCN} < \text{CH}_3\text{COOH} < \text{H}_2\text{O}$
43. If an ideal gas is expanded at constant temperature
(I) The pressure of the gas decreases
(II) Average kinetic energy of the gas decreases
(III) Average kinetic energy of the gas remains same
(IV) The number of molecule of the gas increases
(A) I & II are correct statements
(B) II & III are correct statements
(C) II & IV are incorrect statements
(D) I, II & IV are correct statements
44. For As_2S_3 colloids, which of the following ion will have highest coagulating power?
(A) Mg^{2+}
(B) SO_4^{2-}
(C) PO_4^{3-}
(D) Al^{3+}
45. The product obtained when calcium carbide is strongly heated with N_2 is
(I) $\text{Ca}(\text{CN})_2$ (II) CaCN_2 (III) C (IV) Ca_3N_2
(A) I & II
(B) II & III
(C) I & IV
(D) II only
46. The basic strength order for alkali metal oxide is
(A) $\text{Rb}_2\text{O} > \text{K}_2\text{O} > \text{Li}_2\text{O} > \text{Na}_2\text{O}$
(B) $\text{Li}_2\text{O} < \text{Na}_2\text{O} = \text{K}_2\text{O} < \text{Rb}_2\text{O}$
(C) $\text{Li}_2\text{O} < \text{Na}_2\text{O} < \text{K}_2\text{O} < \text{Rb}_2\text{O}$
(D) $\text{Li}_2\text{O} > \text{Na}_2\text{O} > \text{K}_2\text{O} > \text{Rb}_2\text{O}$
47. Transition metal with highest melting point among the following is
(A) Fe
(B) Mo
(C) Cr
(D) Ru
48. In which of the following ores calcium is not present?
(A) Dolomite
(B) Bauxite
(C) Gypsum
(D) Calcite

49. The drug which is used to relieve pain
 (A) Antipyretic
 (B) Antiallergic
 (C) Antiseptic
 (D) Analgesic
50. The V_2O_5 is used as a catalyst in
 (A) Contact process
 (B) Deacon's process
 (C) Wacker process
 (D) Ziegler –Natta polymerisation

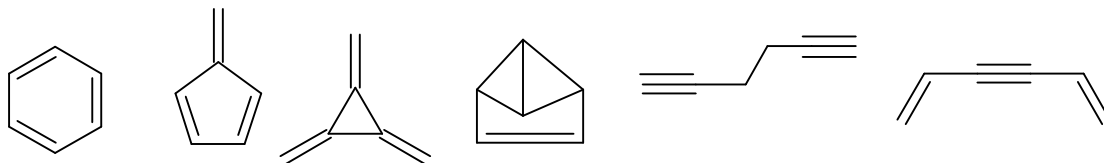
SECTION – B (Numerical Answer Type)

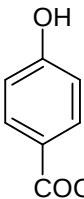
This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXX.XX).

51. What will be the ratio of σ -bond to all $p\pi$ - $d\pi$ bond in SO_3 is
52. How many of the following species will show tautomerism?



53. How many of the following structure of C_6H_6 have all 6 equivalent hydrogen

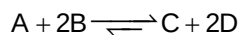


54. 20 ml, 0.1 M  reacts with 20 ml, 0.025 M $Ca(OH)_2$

Solution, what will be pH of final solution [Given : $K_{a_1} = 10^{-5}$, $K_{a_2} = 10^{-9}$]

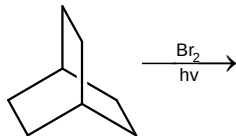
55. If co-ordination number of $[M^{2+}]$ is four, how many moles of acetylacetonato anion per mole of metal ion is/are needed to form stable complex

56. For a homogeneous gaseous reaction



Initial concentration of A is $\frac{2}{3}$ times of B. When equilibrium is achieved concentration of A and B becomes equal. The value of K_p is.....?

57. Total number of monobrominated product (including stereo isomer) is



58. If the half of the face centered atoms are removed, from a ccp type structure. The approximate void efficiency will be (nearest integer)
59. 2 Ampere current is passed into 36 g of pure H_2O at room temperature. The time required (in hour) for complete decomposition of H_2O is ($1F = 96500$)
60. An ideal solution of x and y having vapour pressure in pure liquid state is 36 mm and 25 mm of Hg respectively. If initially 1 mole of both x and y are present in the mixture, then find out the final vapour pressure (in mm) of mixture, when 1 mole of mixture has been vapourised?

Mathematics

PART – C

SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. If z_1 is a root of the equation $z^5 + z^3 + z + 3 = 0$, then
 (A) $|z_1| < 1$
 (B) $|z_1| \geq 1$
 (C) $|z_1| \leq \frac{1}{2}$
 (D) $|z_1| \leq \frac{1}{3}$
62. The equation $e^{|\sin x|} + e^{-|\sin x|} + 4k = 0$ will have exactly four distinct solutions in $[0, 2\pi]$ if
 (A) $k \in \mathbb{R}^+$
 (B) $k \in \left[-\frac{1}{4}, 0\right]$
 (C) $k \in [-5, -4]$
 (D) none of these
63. An ellipse whose foci are (2, 4) and (14, 9) and touches x-axis, then its eccentricity is
 (A) $\frac{\sqrt{13}}{213}$
 (B) $\frac{13}{\sqrt{179}}$
 (C) $\frac{13}{\sqrt{313}}$
 (D) $\frac{1}{13}$
64. If the point (t, t^2) lies inside the triangle formed by the lines $2x + 3y - 1 = 0$, $x + 2y - 3 = 0$ and $5x - 6y - 1 = 0$, then number of integral value of t is
 (A) 0
 (B) 1
 (C) 2
 (D) 3
65. Let $f(x) = \begin{cases} \ln(\{x\}) & ; x \notin \mathbb{I} \\ 0 & ; x \in \mathbb{I} \end{cases}$; (where $\{x\}$ is fractional part function), then the area enclosed by curves $y = f(x)$, $y = f(|x|)$, $x \in [-5, 0]$ and $y = 0$, is
 (A) $5\ln\left(\frac{e}{2}\right)$
 (B) 0
 (C) $5\ln 2$
 (D) 1

66. A dice is rolled 4 times, the numbers appearing on the dice in different throws are listed, such that the largest number appearing in the list is not 4, in how many ways this can be done
(A) 175
(B) 625
(C) 1040
(D) 1121
67. If the quadratic equation $ax^2 - bx + 12 = 0$ where a, b are positive integers not exceeding 10 has roots both greater than 2, then number of possible ordered pair (a, b) is
(A) 1
(B) 2
(C) 4
(D) 8
68. Let $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ be a function which satisfy $f(x)f(y) = f(xy) + 2\left(\frac{1}{x} + \frac{1}{y} + 1\right) \forall x, y > 0$, then the value of $f\left(\frac{1}{2}\right)$ is
(A) -4
(B) 5
(C) $-\frac{8}{3}$
(D) -5
69. If $(\sin \alpha)x^2 - 2x + b - 2 \geq 0 \forall x \in (-\infty, 1]$ and $\alpha \in \left(0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right)$, then the possible real value of b is
(A) 1
(B) 2
(C) 3
(D) 4
70. The number of terms in the expansion $(a + b + c)^6$ which must contain non zero index (power) of 'a'
(A) 15
(B) 21
(C) 6
(D) 12
71. If $L = \lim_{m \rightarrow \infty} \sum_{p=1}^m \frac{p}{2p + m + m^2}$, then the value of $[L]$, where $[.]$ denotes the greatest integer function
(A) 3
(B) 2
(C) 1
(D) none of these
72. Let $ax + by = 40$ is a chord of minimum length of the circle $x^2 + y^2 - 20x - 40y - 229 = 0$ if chord passes through $(5, 15)$, then the value of $\frac{a}{b}$ is
(A) 4
(B) 1
(C) 13
(D) 17

73. Let $g(x)$ be double differentiable function with continuous derivative, defined for $x \in [0, 1]$ satisfying the equation $\int_0^x (\sqrt{1 - (g''(t))^2} - g'(t)) dt = 0 \quad \forall x \in [0, 1]$ such that $g(0) = 0 = g'(0)$, then
- (A) $g(1) > \frac{1}{2}$
 (B) $g(1) < \frac{1}{2}$
 (C) $g(1) = 1$
 (D) none of these
74. The sum of real roots of the equation of the equation $z^5 - 5z^4 + 9z^3 - 9z^2 + 5z - 1 = 0$, is
- (A) 5
 (B) 4
 (C) 3
 (D) 1
75. If $f(x) = \tan^{-1}\left(\frac{3x - x^3}{1 - 3x^2}\right)$ and $g(x) = \frac{\pi\sqrt{3}}{8}|x - \sqrt{3}|$, then number of real values satisfying the equation $f(x) - g(x) = 0$
- (A) 4
 (B) 3
 (C) 2
 (D) 1
76. If $A = \sum_{r=1}^{\infty} \left(\int_0^1 \frac{x^{r-1}}{r} dx \right)$ and $B = \sum_{r=1}^{\infty} \left(\int_0^1 \frac{x^{2r-2}}{2r-1} dx \right)$, then the value of $\frac{3A+B}{B}$ is
- (A) 0
 (B) $\frac{3}{2}$
 (C) 5
 (D) 4
77. While shuffling a pack of 52 playing cards. Unfortunately 4 cards are dropped on the ground it is known that 2 out of 4 are spades. What is the probability that all dropped 4 are spades?
- (A) $\frac{{}^{13}C_2}{{}^{39}C_4}$
 (B) $\frac{{}^{13}C_2}{{}^{52}C_4}$
 (C) $\frac{{}^{13}C_2}{{}^{39}C_2}$
 (D) none of these
78. The number of ways of selecting 4 letters from 3 'a's, 4 'b's and 5 'c's is
- (A) $\frac{{}^8C_2}{2}$
 (B) 6C_4
 (C) $\frac{{}^{12}C_4}{3}$
 (D) ${}^3C_2 + {}^4C_2 + {}^5C_2$

79. The equation of the common tangent to the curves $\left|\frac{z}{2} - i\right| = 1$ and $|z - 1 - 3i| + |z + 1 - 3i| = 2\sqrt{2}$ is
- (A) $z - \bar{z} = 4i$
 (B) $z - \bar{z} = 8i$
 (C) $z + \bar{z} = 2$
 (D) $z + \bar{z} = 4$
80. If the equation $x^2 + (\alpha - 2)x = 3\left(|x| - \frac{1}{3}\right)$ has exactly three distinct real solutions, then which of the following value α can **NOT** take
- (A) 1
 (B) -3
 (C) 3
 (D) 7

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

81. If $\alpha = \tan \frac{\pi}{7}$, $\beta = \tan \frac{2\pi}{7}$, $\gamma = \tan \frac{4\pi}{7}$, then the value of $\frac{|\alpha + \beta + \gamma|}{(\alpha\beta\gamma)^2}$ is equal to
82. If P_1 and P_2 are two complex numbers, then the maximum value of $\frac{P_1\bar{P}_2 + P_2\bar{P}_1 + P_1P_2 + \overline{P_1P_2}}{|P_1||P_2|}$ is equal to
83. If (a, b) satisfies the equation $5a^2 - 12ab - 24 = 0$ and $\sqrt{a^2 + (b+1)^2} + \sqrt{(a-12)^2 + (b-4)^2} = 13$, then $a + b$ is equal to
84. For real value α let $P(\alpha) = 1 + \cos \alpha + i \sin \alpha - \cos 2\alpha - i \sin 2\alpha + \cos 3\alpha + i \sin 3\alpha$ when $i = \sqrt{-1}$ and if number of value of α , $\alpha \in [0, 2\pi)$ for which $P(\alpha) = 0$ is k , then value of $\left(k + \frac{3}{4}\right)$ is
85. If $P = {}^3C_2 C_2 + {}^4C_2 C_2 + {}^5C_2 C_2 + \dots + {}^{42}C_2 C_2$, then the value of $\left\{\frac{P}{10}\right\}$ is (where $\{.\}$ is denote fractional part function)
86. Distinct lines L_1 and L_2 lie in the xy plane intersect at $(0, 0)$, point $P(-1, 4)$ is reflected about the line L_1 to P' and then P' reflected about line L_2 to $P''(4, 1)$ and if equation of L_2 is $ax + by = 0$, then the value of $\left|\frac{b}{a}\right|$ is
87. Let S is the set of all the polynomial $P(x) = ax^3 + bx^2 + cx + d$, where $a, b, c, d \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ if one polynomial $P(x)$ is selected from the set S , then what is the probability that $P(-1) = -9$

88. If line $y = mx + 2$ passes through no such $(\alpha, \beta) \in I$ with $\alpha \in (0, 100]$ for all m such that $m \in \left(\frac{1}{2}, a\right)$. If maximum possible of m is k , then $\frac{1}{k}$ is
89. If $\lim_{x \rightarrow 0} \frac{\cos^2 x - \cos x - e^x \cos x + e^x - \frac{x^3}{2}}{x^n} = L$. When L is non-zero finite $n \in \mathbb{N}$, then the value of (nL) is
90. Each face of a cube is painted either red or blue each with probability $\frac{1}{2}$, the colour of each face is determined independently, if the probability that the painted cube can be placed on horizontal surface so that the four vertical faces are all the same colour is P , then $\frac{1}{P}$ is